

Data

GEMStat
943KUSGS NWIS
163KADES
694KEEA
288KBangladesh
1,807

Tokenize

Harmonization → 20 chemistry parameters + 18 GEE auxiliary features

Input

[CLS]

As

Fe

Eh

PO₄

U

...

elev

NDVI

TWSA

...

WBT

chemistry tokens

auxiliary (satellite) tokens

each token = type embedding + per-parameter value projection

Encoder (N×)

N×

Multi-Head
Self-Attention

Add & Norm

Feed Forward

Add & Norm

Pool

[CLS] pooling → latent $h \in \mathbb{R}^d$

Heads

MGM Reconstruction Head
(masked value prediction)Classifier Head (MLP)
WHO-threshold exceedance

pretraining objective

downstream task